# Document Clustering project

In [29]:
```
# Import necessary libraries
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
```

In [30]:
```
# Step 1: Input text data
# Assume you have a list of documents as follows:
documents = [
    "Love encompasses a range of strong and positive emotional and mental states, from the most sublime virtue or good habit, the deepest interpersonal affectic
    "People can have a profound dedication and immense appreciation for an object, principle, or objective, thereby experiencing a sense of love towards it. For
    "Although the nature or essence of love is a subject of frequent debate, different aspects of the word can be clarified by determining what is not love (ant
    "Friendship is a relationship of mutual affection between people.It is a stronger form of interpersonal bond than an acquaintance or an association, such as
    # Add more documents as needed
```

In [31]:
```
# Step 2: Represent each article as a vector
# Use TF-IDF vectorization for text data
vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(documents)
```

In [32]:
```
# Step 3: Perform k-means clustering
k = 3  # Specify the number of clusters
kmeans = KMeans(n_clusters=k, random_state=42)
kmeans.fit(X)
```

```
C:\Users\B N Rao\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1416: FutureWarning: The default value of `n_init` will change from 10 to 'auto' in 1.
4. Set the value of `n_init` explicitly to suppress the warning
  super()._check_params_vs_input(X, default_n_init=10)
```

Out[32]:

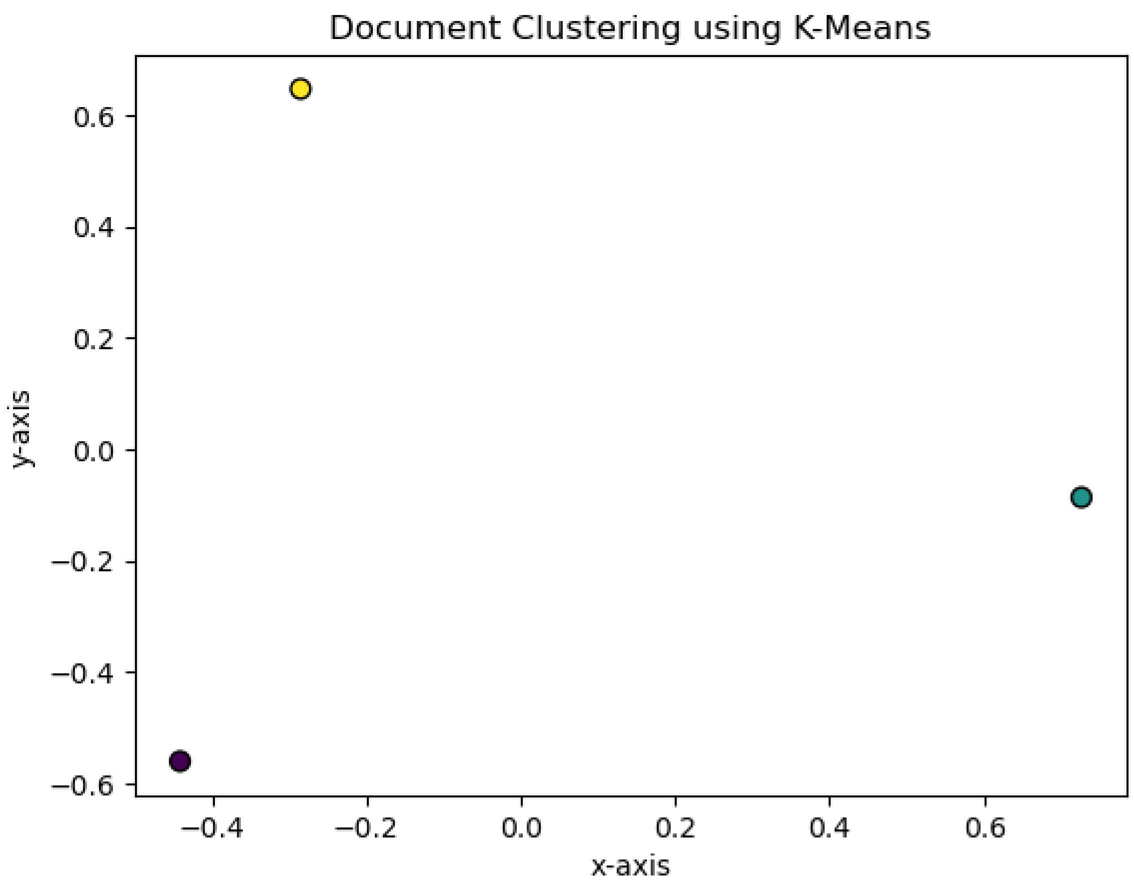
```
▾ KMeans
KMeans(n_clusters=3, random_state=42)
```

In [33]:
```
# Add cluster labels to the original data
labels = kmeans.labels_
df = pd.DataFrame({'Document': documents, 'Cluster': labels})
```

In [34]:
```
# Step 4: Show data output (data visualization)
# Reduce dimensionality for visualization (using PCA)
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X.toarray())
```

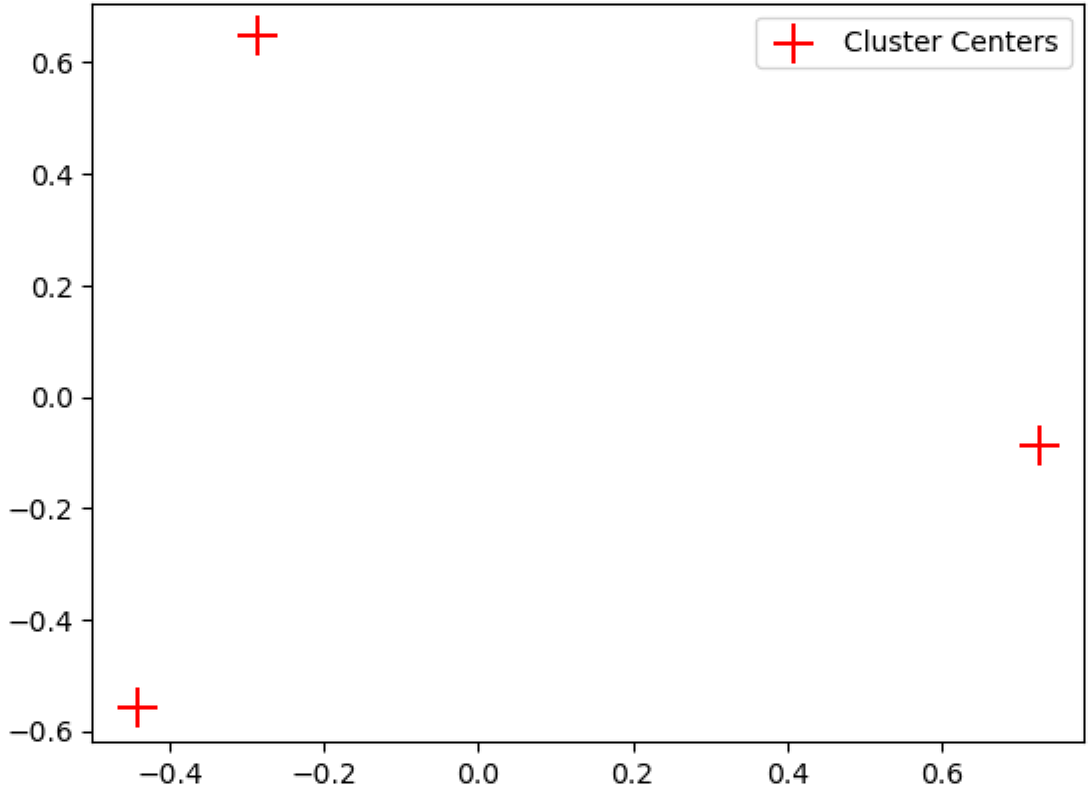
In [37]:
```
# Plot the clusters
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=labels, cmap='viridis', edgecolor='k', s=50)
plt.title('Document Clustering using K-Means')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```

Out[37]:
```
Text(0, 0.5, 'y-axis')
```




In [38]:
```
# Display cluster centers if needed
centers = pca.transform(kmeans.cluster_centers_)
```

```
plt.scatter(centers[:, 0], centers[:, 1], c='red', marker='+', s=200, label='Cluster Centers')
plt.legend()
plt.show()
```




```
In [ ]:
```